

Information on living with diabetes:

-Diabetes causes blindness, kidney failure, heart attacks, stroke and lower limb amputation. Neuropathy is also common — diabetic neuropathy, often manifesting as distal symmetrical polyneuropathy, is among the most frequent complications of both type 1 and type 2 diabetes.

- Measure at least 4-6 times a day
- For some users it can be emotional overwhelming and even cause anxiety.
- The usage of app for diabetes measurements are effective, just motivating the people to use it is tricky. Make it attractive for the users.
- The strongest design principles favored by users in a co-design study of a diabetes-adjacent digital intervention were social support, reduction (cutting the effort needed to reach a target behavior), tunnelling (guided step-by-step flows), praise, rewards, and self-monitoring. With the most-preferred prototype framed around recognizing a user's journey, progress, successes, and failures in a supportive, encouraging way, not a purely metrics-driven one.

Practically, for your N8N build:

an optional validated distress check-in (e.g., short PAID-style questions) that adapts subsequent app tone/support

- Encouraging, progress-framed feedback messages instead of raw pass/fail glucose alerts
- A lightweight peer or "buddy" support loop, even simple as shared encouragement messages
- Automation (N8N is well-suited here) for gentle, well-timed reminders rather than nagging alerts — reducing effort/friction rather than adding more tracking tasks
- Emotional support messages, if wanted.
- Little mascot, easy design, supportive, and helps with easing emotions.

Key basic facts:

It's a single snapshot, not a trend. A finger-prick check only measures glucose at that exact moment — it doesn't show whether the level is rising or falling. That's the main reason people who use insulin often have to test several times a day rather than just once.

Typical frequency: people managing diabetes with insulin commonly check around 3–7 times a day, though this varies by individual, treatment plan, and what their care team recommends.

The alternative: CGM. Continuous glucose monitors use a small sensor worn on the skin that measures glucose in interstitial fluid automatically, often every few minutes,

and show trend arrows (rising/falling and how fast) instead of a single point-in-time number. Some CGMs still require an occasional finger-prick to calibrate.

This suggests two design opportunities — (1) reduce the *number* of manual entries needed (e.g., CGM integration, quick-entry UI) and (2) don't treat glucose logging as a neutral data-entry task — some users may need reassurance or low-friction design because the act itself carries emotional weight for them.

Want me to also cover normal/target blood sugar ranges (fasting vs. post-meal), or move on to CGM specifics?

People want technology, but current tools have real gaps:

In a safety-net hospital study, participants were supportive of a smartphone app for diabetes management and especially interested in tracking blood sugar and communicating with their care team, though technology use specifically for diabetes management was still low, and apps are often not accessible to the populations most vulnerable to poor outcomes. Separately, the "easy to use" factor was found to be the most valuable feature of current self-management apps, while gamification — despite being a useful learning technique — was largely absent from initial training in the devices studied. On CGM specifically, users described it as empowering because it let them access glucose data effortlessly and see trend arrows showing whether glucose was rising or falling and how fast.

What it's like to actually do it (not just theory):

This is worth knowing for your app design:

For some people finger-prick monitoring causes little issue, but for others it causes real **stress and anxiety** — this is a recognized, common reaction, not a rare exception.

Frequent finger sticks over time can also cause practical annoyances: sore or calloused fingertips, and general fatigue with the routine — part of why people find it burdensome to keep doing multiple times a day, every day, indefinitely.

Accuracy is finicky in small ways: the first drop of blood can be contaminated and give a wrong reading; squeezing too hard can dilute the sample with tissue fluid; cold hands reduce blood flow and make it harder to get a usable sample.

So I was thinking about making it interactive game, where you can take advancements. Make it a fun experience to also help with the mental health to fully cover the SDG. What you think?

Sources:

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